Forum:	World Bank
Issue:	Expanding financial inclusion through digital infrastructure in developing states
Author:	Ari Tingis
Position:	Deputy President

Personal Introduction

Dear delegates,

My name is Ari Tingis, and I am currently an IB1 student at St. Catherine's British School. It is my honour and privilege to serve as your Deputy President in the World Bank committee of the first Moraitis MUN.

What started as just a simple curiosity has now bloomed into one of the most enlightening aspects of my academic life. I enrolled in Model UN two years ago, realised that Model UN tied quite closely with my personal interests, and since then I have continually fallen in love with the excitement of momentarily becoming a diplomat. Looking back at fourteen, I never imagined I would possess today's skills in public speaking and clear expression.

In this year's conference, I will be directing the topic "Expanding financial inclusion through digital infrastructure in developing states". This study guide's primary purpose is to provide a base for you to build your knowledge off of while you research and learn more independently on matters expanding further than what is discussed in this guide. The bibliography can also indicate where to source more information from, so please use it as you see fit.

I am here to guide you prior to and throughout the conference. Should you have any questions about the study guide or regarding the conference, please do not hesitate to contact me via email at aritingis121@gmail.com.

I look forward to meeting all of you soon.

Kind regards,

Ari Tingis

Introduction

When we introduce the phrase financial inclusion, we picture a world where all financial services are accessible to all by removing barriers for those excluded, with education and income being the two key pillars for financial inclusion.[1] A significant proportion of adults in less economically developed countries (LEDCs) remain excluded from formal financial services.[2] Being excluded from formal financial services has many negative effects: such populations may struggle to protect themselves in times of crisis, do not have access to resources, and therefore cannot meet fundamental needs.[3] Financial exclusion can also be considered a factor that hinders economic development, which we will expand on later in this guide.

Over the past decade, modern technology and digital infrastructure have become valuable tools that enable financial inclusion for all individuals, including in areas underserved by banks. People can use their mobile phones and the Internet to access financial services independently, without necessarily visiting a physical bank branch. Digital banking and mobile money make it possible to receive salaries, make transactions, open businesses, and participate actively in both regional and global economies where the necessary infrastructure is available.

However, even today, many LEDCs face various obstacles on their way to financial inclusion: insufficient digital infrastructure, low digital literacy rates among the population, and high costs associated with modern technologies prevent millions of people in these nations from benefiting from digital solutions to financial problems.[4] The issue of cybersecurity and data security also requires special attention as systems of weak security can expose individuals to financial fraud and discourage trust in digital financial services.[5] Consequently, individuals who lack access to financial services and find it harder to benefit from economic opportunities are the product of existing inequalities coupled with poor security infrastructure.[6]

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[1] Girón, A., Kazemikhasragh, A., Cicchiello, A. F., & Panetti, E. (2021). Financial Inclusion Measurement in the Least Developed Countries in Asia and Africa. *Journal of the Knowledge Economy*. https://doi.org/10.1007/s13132-021-00773-2

[2] Financial Inclusion | World Bank Group. (2025). In *Worldbank.org*. https://www.worldbank.org/ext/en/topic/financial-sector/financial-inclusion

[3] *The impact of Global Financial Exclusion - WorldRemit*. (n.d.). WorldRemit. https://www.worldremit.com/en/resources/global-financial-inclusion-and-exclusion

[4] Fietz, K., & Lay, J. (2023). Digital Skills in the Global South: Gaps, Needs, and Progress. In *www.giga-hamburg.de*. https://www.giga-hamburg.de/en/publications/giga-focus/digital-skills-in-the-global-south-gaps-needs-and-progress

[5] The role of cybersecurity and data security in the digital economy. (n.d.). In *UNCDF Policy Accelerator*. Retrieved July 21, 2026, from https://policyaccelerator.uncdf.org/all/brief-cybersecurity-digital-economy

[6] *Financial inclusion | World Bank Group*. (2025). Worldbank.Org. https://www.worldbank.org/ext/en/topic/financial-sector/financial-inclusion

Several disparities in digital and financial infrastructure can contribute to a divergent development gap between higher-income economies and the Global South.[7] In other words, developing nations with limited technological capacity may face greater difficulties integrating into an increasingly digital global economy.[8] Therefore, expanding financial inclusion through digital innovation is crucial not only for increasing access to financial services, but also for promoting broader economic participation and ensuring that the benefits of digitalisation are accessible to developing populations in LEDCs.

Key Terms and Definitions

Digital Infrastructure

"Digital Infrastructure is the technological foundation that businesses, the government, and society rely on. Some examples are: computers, smartphones, applications, cloud computing services, and cybersecurity systems, making it a sector that is deeply involved in our economy."[9]

Cloud Computing

Cloud computing can be defined as "the use of services, computer programs, etc. that are on the internet rather than ones that you buy and put on your computer".[10] Cloud Computing is important in developing countries because it removes the need for expensive physical hardware and allows local businesses, schools, and governments to use advanced technology at a low cost.[11]

Financial Inclusion

"Financial inclusion refers to the process of ensuring that individuals and businesses, regardless of their income levels or social status, have access to useful and affordable financial products and services."[12]

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[7] *International investment in the digital economy: A toolkit for policymakers* |. (2026, January 16). UN Trade and Development (UNCTAD). https://unctad.org/publication/international-investment-digital-economy-toolkit-policymakers

[8] Berk Tuttup. (2024, August 31). *What is the global south?* Foreign Analysis. https://foreignanalysis.com/what-is-the-global-south/

[9] Neos Networks. (2025, October 15). *What is digital infrastructure?* https://neosnetworks.com/resources/blog/what-is-digital-infrastructure/

[10] Cambridge Dictionary. (2025, January 8). *Cloud computing*. @CambridgeWords. https://dictionary.cambridge.org/us/dictionary/english/cloud-computing

[11] *How cloud computing is contributing to the exponential growth of tech in emerging markets*. (2018, February 13). Seedstars. https://www.seedstars.com/content-hub/life/how-cloud-computing-contributing-exponential-growth-tech-emerging-markets/

[12] *Financial inclusion - Definition and Explanation*. (2024, November 21). The Oxford Review - or Briefings. https://oxford-review.com/the-oxford-review-dei-diversity-equity-and-inclusion-dictionary/financial-inclusion-definition-and-explanation/

Mobile Banking

"Mobile banking is a service that allows users to conduct financial transactions using a mobile device, such as a smartphone or tablet. Mobile banking features can be as simple as a bank sending fraud or usage activity to a client's cell phone or as complex as a client paying bills or sending money abroad.[13] In developing countries, mobile banking is an entry point for formal finance for individuals who lack access to physical bank branches but own basic smartphones.

Digital Transactions

"A digital transaction refers to one done electronically. It can involve transferring funds, exchanging information, or making a contract without needing a physical presence or paper-based documents,"[14] which serves as a useful model in this context because reduced reliance on cash and paper documentation reduces cost and increases efficiency,[15] thereby making digital transactions more accessible for populations in resource-constrained locations.

Financial Literacy

"Financial literacy is a combination of awareness, knowledge, skill, attitude and behaviour necessary to make sound financial decisions and ultimately achieve individual financial well-being,"[16] in other words, the ability to effectively navigate, comprehend and use financial tools. Embedding financial literacy into financial tools empowers users in developing states to confidently use them, therefore improving inclusion.

Know Your Customer (KYC)

"Know Your Customer (KYC) standards are designed to protect financial institutions against fraud, corruption, money laundering and terrorist financing."[17] For example, institutions may require a proof of Address, Proof of Identification or biometric verification to verify client identities.[18]

Fintech

"Finance and technology. These are the terms behind the word 'fintech', a name used to describe activities involving the use of technology and innovation for the provision of services and the design

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[13] *Financial inclusion - Definition and Explanation*. (2024, November 21). The Oxford Review - or Briefings. https://oxford-review.com/the-oxford-review-dei-diversity-equity-and-inclusion-dictionary/financial-inclusion-definition-and-explanation/

[14] *What Is a Digital Transaction?* (n.d.). Investopedia. https://www.investopedia.com/terms/d/digital-transaction.asp

[15] Admin. 'Top 6 Benefits of Digital Payments for Businesses | Sycurio'. *Sycurio.Com*, Sycurio Limited, 11 July 2024, https://sycurio.com/blog/top-six-benefits-of-digital-payments.

[16] *Financial-Literacy*. (2026). Bankofgreece.Gr. https://www.bankofgreece.gr/en/the-bank/social-responsibility/financial-literacy

[17] *What Is a Digital Transaction?* (n.d.). Investopedia. https://www.investopedia.com/terms/d/digital-transaction.asp

[18] Bluemel, J. (2022, March 8). *What is KYC? Know your customer verification explained*. IDnow. https://idnow.io/what-is-kyc/

of financial products."[19] Fintech startups in developing states are outpacing traditional banking models by designing affordable products tailored to local needs and behaviours.

Less Economically Developed Countries (LEDCs)

"LEDCs or Less Economically Developed Countries are countries with a low standard of living and a much lower GDP,"[20] compared to more economically developed countries (MEDCs). Formal financial penetration in LEDCs is generally low.[21]

Central Bank

"A central bank is a public institution that manages the currency of a country or group of countries and controls the money supply."[22] Some central banks in developing states are exploring or issuing digital currencies to stabilise economies and expand access to secure digital money.[23]

Payment Infrastructure

Payment infrastructure can be defined as a "complex network of technologies, financial institutions, and compliance standards that modern payments depend on"[24]. In other words, they are an integrated financial network which enables money to move securely from a payer to a payee.

Commercial Bank

"Commercial banks provide basic banking services and products to the general public, both individual consumers and businesses, and they also provide their business clients with merchant services."[25] To support financial inclusion, commercial banks can partner with fintechs and leverage agent networks to extend their reach beyond urban centres to serve rural and informal sector clients.[26]

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[19] *What is fintech or financial technology?* (2024, March 15). EAE Barcelona Business School. https://www.eaebarcelona.com/en/blog/what-is-fintech

[20] The development gap - CCEA. (n.d.). In *BBC Bitesize*. Retrieved July 21, 2026, from https://www.bbc.co.uk/bitesize/articles/zwf63qt#zjytb7h

[21] 'Bridging the Informal-Formal Divide in Financial Services'. *Www.Uncdf.Org*, https://www.uncdf.org/article/8190/bridging-the-informal-formal-divide-in-financial-services.

[22] European Central Bank. (2021). What is a central bank? In the European *Central Bank*. https://www.ecb.europa.eu/ecb-and-you/explainers/tell-me/html/what-is-a-central-bank.en.html

[23] Huang, Weihang, and Breidbach Christoph F. 'Central Bank Digital Currencies: What Is Known and yet to Be Known?'. *The Journal of Strategic Information Systems*, vol. 35, no. 3, Sept. 2026, p. 101978, https://doi.org/10.1016/j.jsis.2026.101978.

[24] 'Payment Infrastructure: Definition, How It Works & Key Components'. *Decta.Com*, 2025, https://www.decta.com/company/media/payment-infrastructure.

[25] Understanding Commercial Banks. (n.d.). In *Investopedia*. Retrieved July 21, 2026, from https://www.investopedia.com/terms/c/commercialbank.asp#toc-what-is-a-commercial-bank

[26] *Prosightfa.Org*, 2026, https://www.prosightfa.org/insights/fintechs-can-be-a-community-banks-ally-in-a-competitive-digital-world/.

Capital Markets

A capital market can be defined as "a financial middle ground where asset exchanges between investors and sellers who need long-term funds to grow or build projects occur."[27] They are usually managed by entities in the financial sector[28] and digitizing capital market access enables small-scale investors and entrepreneurs in developing states to participate in growth opportunities previously reserved for large players.

Informal Finance

"Informal finance refers to financial activities that occur outside regulatory authorities, relying heavily on personal relationships between providers and users. This sector includes various institutions like moneylenders and savings associations and is particularly active in areas lacking formal financial services."[29]

Formal Finance

"Financial services provided by regulated entities such as banks, non-bank financial institutions and capital markets, typically governed by law and requiring documentation and collateral"[30] and it enables states to bring users under regulatory consumer protection.

Microfinance

"Microfinance provides small loans and financial services to underserved individuals or communities to help them achieve self-sufficiency and economic independence",[31] and integrating microfinance in mobile platforms will enable them to expand their reach and impact to foster lending at scale across different regions.

Digital Public Infrastructure (DPI)

"Digital Public Infrastructure (DPI) refers to the foundational digital systems and technologies that support the delivery of public and private services to individuals and organizations."[32] DPI fundamentally acts as the backbone for financial inclusion in developing states by enabling secure,

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[27] Hayes, Adam. 'Capital Markets: What You Should Know'. *Investopedia*, 30 Aug. 2021, https://www.investopedia.com/terms/c/capitalmarkets.asp.

[28] Wikipedia Contributors. 'Capital Market'. *Wikipedia*, Wikimedia Foundation, 3 Nov. 2019, https://en.wikipedia.org/wiki/Capital_market.

[29] Andreosso-O'Callaghan, B., & Gottwald, J. (2014). How risky is China's red capitalism? Restructuring in the Chinese financial services sector. *The Globalization of Chinese Business*, 149–167. https://doi.org/10.1533/9781780634494.2.149

[30] *Formal and Informal Finance in Developing Economies | Finance | Banking, Finance and Investment | Social sciences | Topics | Nature Index*. (2025). NatureIndex; Nature Index. https://www.nature.com/nature-index/topics/l4/formal-and-informal-finance-in-developing-economies

[31] Kagan, J. (2024). Microfinance Definition: Benefits, History, and How It Works. In *Investopedia*. https://www.investopedia.com/terms/m/microfinance.asp

[32] *Digital Public infrastructure*. (n.d.). ITU. https://www.itu.int/en/ITU-D/Technology/Pages/dpi.aspx

accessible and compatible delivery of financial services to every citizen regardless of their income or location.

Open Banking

"Open banking allows financial institutions to securely share customer data with third-party providers to offer additional financial services to customers."[33]

Blockchain Infrastructure

"Blockchain is a digital database that records and stores data across a network of computers"[34]. therefore reducing the need for multiple ledgers and enhances transparency in cross-border payments which is a key barrier to inclusion in developing states.[35]

Regulatory Sandboxes

"A regulatory sandbox constitutes a regulatory regime that provides financial technology firms (FinTechs) a controlled environment to test financial propositions for a specified period and while formally engaging with the Regulator."[36]

Background Information

Historical Background

The foundations of the modern financial system developed during the 18th and 19th centuries, alongside the rise of modern economic thought following Adam Smith's The Wealth of Nations (1776). Smith's core insights in light of his publication, such as the 'division of labour' and his 'invisible hand theory', essentially the concept of an invisible capitalist hand emerging from the competing claims of self-interest, echo loudly in later pivotal figures whose theories and principles formed a modern 'school of economic thought'.[37] You could argue that Adam Smith shaped economics as a social science: he treated the economy as an observable and analyzable system[38] and embedded economic behaviour into societal context in his publication, 'The Theory of Moral Sentiments', and studied how individual actions aggregate into national phenomena.[39] Not to

[33] Team, I. (2026, July 18). *Open Banking: definition, how it works, and risks*. Investopedia. https://www.investopedia.com/terms/o/open-banking.asp

[34] *What is blockchain? L fidelity*. (n.d.). Www.Fidelity.Com. Retrieved 19 September 2026, from https://www.fidelity.com/learning-center/trading-investing/what-is-blockchain

[35] *Financial Inclusiveness Report*. https://www.unescwa.org/sites/default/files/pubs/pdf/financial-inclusiveness-report-english.pdf.

[36] *Regulatory-sandbox*. (2019). Bankofgreece.Gr. https://www.bankofgreece.gr/en/main-tasks/supervision/regulatory-sandbox

[37] Butler, C. S., & Taverniers, M. (2008). Introduction. *Linguistics*, *46*(4). https://doi.org/10.1515/ling.2008.023

[38] University of Glasgow. (n.d.). *University of Glasgow - explore - Adam Smith - life, work and legacy - legacy - economy, politics and society*. Www.Gla.Ac.Uk. Retrieved 18 September 2026, from https://www.gla.ac.uk/explore/adamsmith300/lifeworkandlegacy/legacy/economypoliticsandsociety/

[39] *The theory of moral sentiments*. (n.d.). Panmure House. Retrieved 18 September 2026, from https://www.panmurehouse.org/adam-smith/works/the-theory-of-moral-sentiments/

mention that his belief that humans are not perfectly balanced and rational creatures[40] is projected into modern fintech designs which guide good decisions through nudges, fraud alerts and literacy tools and take into account subconscious inclinations when making choices.[41] Despite Smith's foundational outlooks, for much of modern history, having access to financial services depended on having a physical bank nearby, a sufficient income, and the documentation needed to open an account.[42] This created barriers for people living in rural areas and low-income communities in less economically developed countries (LEDCs), where banks were often concentrated in urban centres.[43] As a result, large parts of the population relied on informal methods of saving, borrowing and transferring money.[44]

As economies became more and more industrialized and international trade expanded, formal banking systems continued to develop, yet financial exclusion remained significant in developing states.[45] Over time, the way money worked changed too: first, many countries used the gold standard and tied their currency value to the amount of gold they possessed and where paper money could be exchanged for gold;[46] then after World War II, the Bretton Woods system tied most currencies to the U.S. dollar and then the dollar to gold in order to prevent post war currency depreciation. After the collapse of the Bretton Woods monetary arrangements, most major currencies operated as fiat money,[47] meaning its value depends on trust in governments and central banks rather than physical assets, which was important to note because if people don't trust their currency or banks, they may be less likely to use digital finance even when it is available.

The introduction of microfinance, especially through the the Grameen microcredit project in Bangladesh in the 1970s (later established as a bank in 1983) helped to address some of these issues by providing financial services to populations in rural areas through a system of microfinance in

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[40] alden110. 'Adam Smith: The Theory of Moral Sentiments – Notes From the North Country'. *Notesfromthenorthcountry.Com*, 8 July 2022, https://notesfromthenorthcountry.com/2022/07/adam-smith-the-theory-of-moral-sentiments/.

[41] admin. (2025, August 20). *Fintech UX design: Patterns that build trust and credibility*. Phenomenon Studio. Phenomenon. https://phenomenonstudio.com/article/fintech-ux-design-patterns-that-build-trust-and-credibility/

[42] *Financial access*. (n.d.). World Bank. Retrieved 19 September 2026, from https://www.worldbank.org/en/publication/gfdr/gfdr-2016/background/financial-access

[43] Chandavarkar, A. G. (n.d.). *The financial pull of urban areas in LDCs*. https://doi.org/10.5089/9781616353612.022.a008

[44] SINGH, K. (2019, July 22). *50 Years of Bank Nationalization: A peek into social and development banking - madhyam*. Madhyam. https://www.madhyam.org.in/50-years-of-bank-nationalization-a-peek-into-social-and-development-banking/

[45] Yasir, Z. M., Rafaqat, S., Shaukat, S., Ali, M., & Abbas, S. H. (2025). Financial inclusion as a key driver for economic growth in developing countries. *Journal of Social Sciences Advancement*, *6*(4), 87–92. https://doi.org/10.52223/jssa25-060409

[46] 'The Evolution of Central Banks'. *Hoover Institution*, https://www.hoover.org/research/evolution-central-banks.

[47]Chen, James. 'Fiat Money: What It Is, How It Works, Example, Pros & Cons'. *Investopedia*, 2 July 2024, https://www.investopedia.com/terms/f/fiatmoney.asp.

which small loans are provided to people where access to a traditional physical bank is difficult. This became a foundational precedent for today's digital financial infrastructure, because if small-scale credit could be decentralised, then mobile networks and instant payment systems who sought to bypass physical constraints could scale that same logic. Following the Great Depression, stringent banking rules were introduced to ensure stability; but by the 1970s, high inflation, slow growth, and pressure from banks led governments to see those rules as obstacles, paving the way for deregulation. Therefore, during the 1980s and 1990s, many countries relaxed rules on banks to boost economic growth and international competition, allowing them to grow bigger and operate across borders.[48] This brought more investment and innovation, but also led to financial crises in Asia (1997) and globally (2008)[49], reiterating the need for consumer protection and proper regulation frameworks, and at the same time, payment systems evolved from slow, paper-based methods (such as checks) to real-time mobile platforms like M-Pesa in Kenya or UPI in India,[50] which enable rapid digital transfers; however, UPI ordinarily links to bank accounts, unlike some mobile-money services. This shift from expanding financial markets to expanding access to financial services helped lay the foundation for contemporary approaches to financial inclusion. Today, developing states increasingly use digital infrastructure, including mobile networks, digital identification systems, and instant payment platforms to overcome some of the barriers that previously excluded individuals from formal financial systems and to make financial services more accessible and usable.

[48] McCauley, R. N., McGuire, P., & Wooldridge, P. (2021, September 20). *Seven decades of international banking*. Bank for International Settlements. https://www.bis.org/publications/qr-202109/seven-decades-international-banking

[49] cchanse. 'Asia and the Global Financial Crisis: Conference Summary'. *SF Fed*, San Francisco Fed, 15 Mar. 2010, https://www.frbsf.org/research-and-insights/publications/economic-letter/2010/03/asia-global-financial-crisis-conference-summary/.

[50] Morris, Julian. 'Digital Payments and Financial Inclusion - International Center for Law & Economics'. *International Center for Law & Economics*, 2024, https://laweconcenter.org/resources/digital-payments-and-financial-inclusion/.

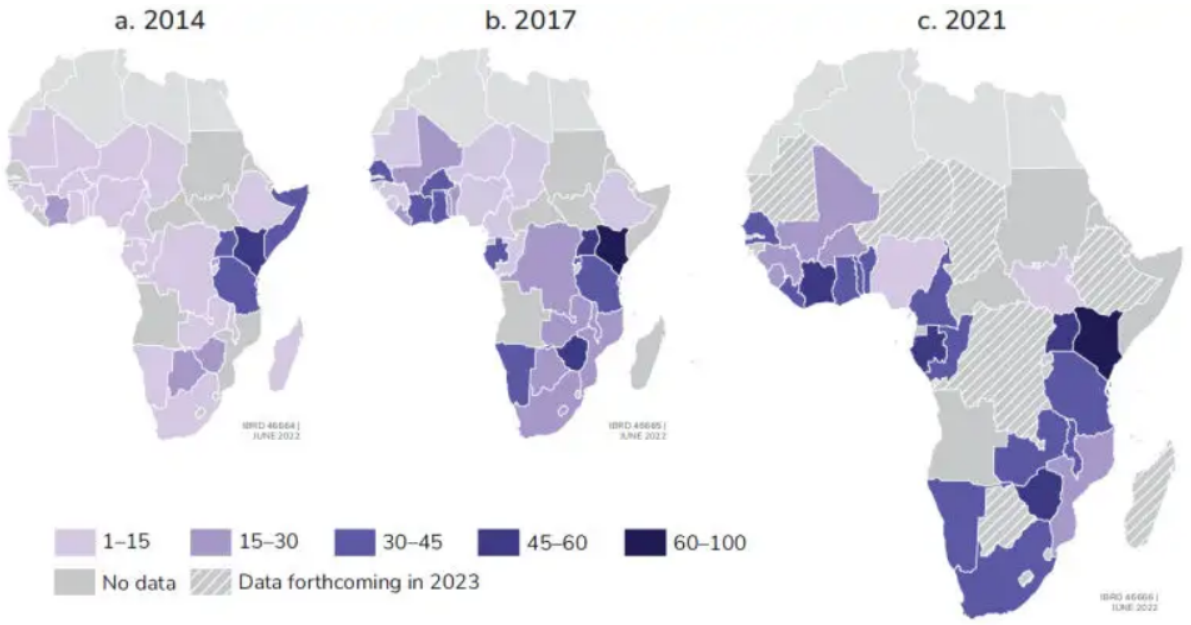


Figure 3: A*dults with a mobile money account (%)*[51]

How do formal banking systems work?

Formal banking systems are made up of regulated institutions, mainly comprising commercial and central banks that accept deposits, lend money, and manage payments under government supervision.[52] When you deposit money in a bank, instead of remaining stoic there, the bank may use part of its funding to make loans to businesses or individuals while meeting liquidity and capital requirements[53], earning interest that helps repay the lender later.. Central banks influence monetary conditions and, in many countries, help supervise banks or support financial stability, although deposit protection and supervisory responsibilities vary by country.[54] Payments between people or companies move through secure systems (like wire transfers or card networks) that banks connect to, making sure money goes where it's supposed to without being lost or stolen.[55] Being part of the formal banking system means members enjoy several notable benefits. Firstly, deposits may be protected by

[51] *Latest Global Findex Data Chart 10 Years of Progress in Financial Inclusion | eTrade for all*. (2023, February 2). Etradeforall.org. https://etradeforall.org/news/latest-global-findex-data-chart-10-years-progress-financial-inclusion

[52] Lee, Kyle. 'Banking System: Definition & Types - Video & Lesson Transcript | Study.Com'. *Study.Com*, 2019, https://study.com/academy/lesson/banking-system-definition-types.html.

[53] *Understanding Central Banking and Monetary Policy - Maseconomics*. 23 Nov. 2022, https://maseconomics.com/understanding-central-banking-and-monetary-policy/.

[54] *Understanding Central Banking and Monetary Policy - Maseconomics*. 23 Nov. 2022, https://maseconomics.com/understanding-central-banking-and-monetary-policy/.

[55] *Payments | Definition, Payment Systems, Settlement vs. Finality*. (2026). AFP. https://www.financialprofessionals.org/glossary/payments

deposit-insurance schemes, where such schemes exist and within their coverage limits. As a result, eligible depositors may benefit from protection if their bank fails.[56] Furthermore, the nature of cash-based economies renders investment highly improbable, yet with money stored in a bank account, banks may use deposits to finance lending and other investments, which are a share of a company's profits paid out regularly to shareholders,[57] and bonds-debt certificates from a loan to a corporation or government which pays regular interest and the return of the initial amount in due course.[58] Finally, when more people join the formal system, it augments the whole economy: governments and central banks can track economic activity more accurately and respond better to aggregate problems like inflation or unemployment.[59]

Effects of Inclusion in Formal Banking/Financial Systems

Formal banking accounts provide many benefits past holding your money and assets. First and foremost, possessing a proper bank account consolidates your entire spending record in one secure location, serving as verified, permanent proof of your financial activity. This service provides households and small firms access to capital through transaction history, which may in turn improve access to finance as the rising domestic savings base funds investments more reliably.[60] With people having formal accounts, governments can make payments of social services directly into their bank accounts,[61] which is benevolent because direct transfers can ensure that subsidies, pensions, or emergency aid reach intended recipients without intermediaries and transaction records allow for better targeting and monitoring of welfare programmes.[62]

Financially included populations can also benefit from faster post crisis responses and minimise their livelihood disruption since emergency cash and insurance payouts can be transferred digitally to those most affected. Another benefit of formal inclusion is that formal savings mechanisms allow individuals to accumulate assets safely and therefore plan for future expenses,[63] breaking vicious cycles of hand to mouth consumption. Inclusion further reduces reliance on unregulated financial

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[56] *Economic issues no. 9--Protecting bank deposits*. (2020). Imf.Org. https://www.imf.org/EXTERNAL/PUBS/FT/ISSUES9/INDEX.HTM

[57] Hayes, A. (2025, April 25). *Dividend definition*. Investopedia. https://www.investopedia.com/terms/d/dividend.asp

[58] Kat Tretina. (2025, January 28). *What are bonds, and how do you invest in them?* Yahoo! Finance. Yahoo Finance. https://finance.yahoo.com/personal-finance/investing/article/what-are-bonds-175919240.html

[59] *How do central banks typically control inflation?* (n.d.). Www.Stlouisfed.Org. Retrieved 19 September 2026, from https://www.stlouisfed.org/on-the-economy/2016/november/how-central-banks-typically-control-inflation

[60] *SAVINGS ARE as IMPORTANT as CREDIT: DEPOSIT SERVICES for the POOR*. 2002, https://documents1.worldbank.org/curated/en/104331468322452634/pdf/341840ENGLISH0DonorBrief104.pdf.

[61] Citizensinformation.ie. 'How Social Welfare Payments Are Paid'. *Www.Citizensinformation.Ie*, https://www.citizensinformation.ie/en/social-welfare/irish-social-welfare-system/claiming-a-social-welfare-payment/how-social-welfare-payments-are-paid/.

[62] 'The Potential Advantages of DCTs'. *EdClinical*, 2026, https://www.clinicalleader.com/doc/the-potential-advantages-of-dcts-0001.

[63] FINCA Webmaster. 'Fewer Poor People Are Saving: What Does This Mean for Microfinance and Mobile Money?'. *FINCA International*, 12 Sept. 2018, https://finca.org/fewer-poor-people-are-saving.

markets that charge unreasonable rates or employ coercive practices, bringing economic activity into supervised channels with consumer protections. [64]

Causes of Financial Exclusion in LEDCs

Infrastructure Barriers

The absence of financial infrastructure in rural areas is a heavy obstacle for financial inclusion, particularly in LEDCs. Infrastructure barriers include a lack of reliable electricity, mobile network coverage, or affordable internet access needed to use digital financial services.[65] These barriers can directly contribute to financial exclusion because populations living in poorly-connected rural areas may be unable to access or consistently use formal financial services leaving them outside increasingly digital formal financial systems.[66] These barriers exist because governments in developing states may underinvest in rural telecom or power grids because they're expensive to build and serve sparse populations, as extending infrastructure to remote and sparsely populated areas can require substantial investment while creating limited short-term economic returns. Private firms have a tendency to not invest either as they assume that they cannot make substantial profit from low-income users.[67] More recent GSMA data indicate that Malawi reached 87% 4G population coverage in 2025, while around 80% of the population still remained offline despite living within mobile broadband coverage, which suggests that financial exclusion is increasingly shaped along with affordability, digital skills instead of infrastructure deficits.[68]

Documentation Barriers

Documentation barriers primarily prevent people from reaching financial inclusion and fully accessing digital government and public services, because a lack of official ID, proof of address or income documents conflict with rigid KYC compliance standards which are mandatory to open a bank account. Millions of individuals in LEDCs are born in geographically disparate regions where births are never officially registered, leaving them without basic birth certificates, and weak infrastructure makes it practically unjustifiable to obtain an official passport or identity card. Over 84

[64] *Financial consumer protection | digital finance inclusion*. (2017). Worldbank.Org. https://digitalfinance.worldbank.org/topics/financial-consumer-protection

[65] 'Infrastructural and Regulatory Barriers | Digital Finance Inclusion'. *Worldbank.Org*, 2020, https://digitalfinance.worldbank.org/topics/gender/infrastructural-and-regulatory-barriers.

[66] Group, W. B. (2024, July 30). *Connecting for inclusion: Broadband access for all*. World Bank Group. World Bank. https://www.worldbank.org/en/topic/digital/brief/connecting-for-inclusion-broadband-access-for-all

[67] Lavrutich, Maria, et al. 'Transmission Investment Under Uncertainty: Reconciling Private and Public Incentives'. *European Journal of Operational Research*, vol. 304, no. 3, Apr. 2022, pp. 1167–88, https://doi.org/10.1016/j.ejor.2022.04.038.

[68] Newsroom. (2026, August 20). *80% Of malawians remain offline despite coverage – new GSMA report highlights path to inclusive digital access and MWK 1.1 trillion growth*. Newsroom. https://www.gsma.com/newsroom/press-release/80-of-malawians-remain-offline-despite-coverage-new-gsma-report-highlights-path-to-inclusive-digital-access-and-mwk-1-1-trillion-growth/

million displaced people globally lack recognised identification,[69] which can prevent them from meeting bank identification requirements even if they have phones and income. For example, an individual residing in a rural location may have to travel long distances on underdeveloped roads to obtain such documents even though many people in LEDCs work in the informal sector (e.g., subsistence farming, street-sellers, day labour), where they are paid in cash and do not possess formalised payslips, tax returns, or employment contracts. Therefore they have little incentive[70] to obtain a formal document to register their work under the fear of government taxes outweighing the benefit of using a traditional bank, a socioeconomic reality that is yet to match the documentation standards to be integrated into digital finance.[71, 72]

Social barriers / Gender Inequality

The full potential of financial inclusion continues to be inordinately slowed down by systemic obstacles like social limitations, creating invisible barriers particularly for women.[73] In many developing societies, traditional cultural norms have assigned men to manage the household's finances and technology[74], implying that women may have accounts but do not have autonomy to make decisions regarding their usage, and even when women own mobile phones or bank accounts, decision-making often rests with male relatives, restricting their ability to save, borrow, or transact without permission; this can be referred to as the gap between access and agency.[75] However, in light of recent years, the gender gap on financial account ownership has narrowed and, in 2021, stood at 4 percentage points globally, with 78 percent of men and 74 percent of women owning financial

[69] 'Leveraging Digital ID and E-KYC for the Financial Inclusion of Forcibly Displaced Persons: Risks and Opportunities - Digital Frontiers Institute'. *Digital Frontiers Institute*, 26 July 2022, https://digitalfrontiersinstitute.org/resource/leveraging-digital-id-and-e-kyc-for-the-financial-inclusion-of-forcibly-displaced-persons-risks-and-opportunities/.

[70] Baafra, Ama, and Ritam Chaurey. *Formal Sector Incentives and Informality CDEP-CGEG WORKING PAPER SERIES*. 2017, https://cdep.sipa.columbia.edu/sites/cdep.sipa.columbia.edu/files/content/WP40Abeberese.pdf.

[71] 'Formal and Informal Economies'. *BBC Bitesize*, 2021, https://www.bbc.co.uk/bitesize/guides/z3spj6f/revision/1.

[72] '2 Defining the Informal Sector'. *Core-Econ.Org*, 2018, https://books.core-econ.org/insights/informality-and-the-capitalist-economy/02-defining-the-informal-sector.html.

[73] Otitoju, Moradeyo Adebanjo, et al. 'The Challenges of Financial Inclusion in Developing Countries: A Review'. *International Journal of Multidisciplinary Research and Growth Evaluation*, vol. 6, no. 6, 2025, pp. 968–73, https://doi.org/10.54660/.ijmrge.2025.6.6.968-973.

[74] 'Cultural Determinants of Household Decision Making Among the Nandi Community in Kenya - International Journal of Research and Innovation in Social Science'. *International Journal of Research and Innovation in Social Science*, 2024, https://rsisinternational.org/journals/ijriss/articles/cultural-determinants-of-household-decision-making-among-the-nandi-community-in-kenya/.

[75] Salman, Arisha, and Gayatri Murthy. 'From Access to Agency: The Next Chapter in Women's Financial Inclusion'. *CGAP*, 12 Aug. 2025, https://www.cgap.org/blog/access-to-agency-next-chapter-in-womens-financial-inclusion.

accounts.[76] Nonetheless, this statistic fails to indicate the owner or true user of the account, and given that some social norms value men as the controllers, women still do not have equal access to financial services.

Educational Barriers

The inability of populations in developing states to access and integrate finance into their lives is linked to educational constraints. The low levels of literacy and numeracy mean that individuals are unable to thoroughly understand how to use mobile money transfer technologies[77], read terms and conditions, or even identify scam messages, and even people in rural areas without formal education may own mobile devices, but do not have the necessary skills to operate and navigate banking applications safely.[78] For instance, research indicates that financial literacy alone accounts for more than 22 percent of the difference in mobile money usage in rural and urban areas in such countries as Tanzania, deriving from observable geographically-correlated income disparities, where higher income users convert digital knowledge into adoption more articulately in comparison to marginalised economic groups in rural locales due to a slower rate of integration.[79] Without basic education including reading, math, or digital skills, users are more susceptible to mistakes or errors, making the poor and least educated segment of society disengage from the system despite having accounts opened.

Effects of Financial Exclusion in LEDCs

Weak Local Capital Markets and Tax Collection

Local capital markets are domestic financial systems where businesses and governments can raise long-term funds by selling bonds or shares,[8081] and when large segments of the population or small businesses operate outside formal banking and digital payments, they cannot generate the extensive transaction records and credit histories necessary for vibrant domestic capital markets to develop

[76] Klapper, Leora, et al. *Women and Financial Inclusion the Global Findex Database 2021*. https://thedocs.worldbank.org/en/doc/0254f85d1ccceceb6e72362a71eb600e4-0430062023/original/Findex2021-GenderBrief-030823.pdf.

[77] Adel, Nabil. 'The Impact of Digital Literacy and Technology Adoption on Financial Inclusion in Africa, Asia, and Latin America'. *Heliyon*, vol. 10, no. 24, Dec. 2024, pp. e40951–51, https://doi.org/10.1016/j.heliyon.2024.e40951.

[78] Menberu, Abebe Walle. 'Technology-Mediated Financial Education in Developing Countries: A Systematic Literature Review'. *Cogent Business & Management*, vol. 11, no. 1, Feb. 2024, https://doi.org/10.1080/23311975.2023.2294879.

[79] Mwaseba, Steven Lee, et al. 'Bridging the Digital Financial Divide: The Role of Financial Literacy in Rural–Urban Disparities in Mobile Money Account Ownership in Tanzania'. *Cogent Economics & Finance*, vol. 14, no. 1, Feb. 2026, https://doi.org/10.1080/23322039.2026.2624866.

[80] 'Capital Market'. *Wikipedia*, 15 July 2026, https://en.wikipedia.org/w/index.php?title=Capital_market&oldid=1364279674.

[81] dusted. 'Including Underserved Communities Through Financial Services'. *British International Investment*, CDC Group, 11 Aug. 2017, https://www.bii.co.uk/en/emerging-markets-investment/including-underserved-communities-through-financial-services/.

because operations in cash leave no digital footprint. By default, vulnerable small and medium enterprises (SMEs), farmers, and female-owned businesses face persistent difficulties in obtaining affordable long-term financing[82] beyond expensive bank loans[83] or informal lenders, which in turn constrains their ability to expand operations or invest productivity-enhancing mechanisms like more workers or machinery, consequently limiting job creation and broader economic growth by reducing productive investment and the capacity of businesses to scale[84]. These challenges are exacerbated and intensified by weak local capital markets, which restrict the availability of affordable, long-term sources of finance and therefore limit the ability of smaller businesses to contribute fully to economic development.[85]

In parallel, the prevalence of cash-based and informal economic activity renders income and transactions rather invisible to tax authorities, making it more difficult for the government to identify its taxpayers, underline what needs investment, and collect revenue to its maximum efficiency, which results in chronically underfunded public services, such as: roads, electricity, connectivity, education, healthcare, digital identity systems, financial regulation, and consumer protection frameworks,[86] all of which are prerequisites for a proper foundation to financial inclusion. This reduction in tax revenue consequently weakens the government's fiscal capacity, meaning its ability to raise and allocate sufficient funds for public expenditure and investment.[87] As a result, governments may become more reliant on borrowing to finance essential expenditure, while having fewer resources available for public investment. In this case, social safety nets are left inadequately funded, limiting the state's ability to support vulnerable populations during economic shocks. In turn, insufficient investment in infrastructure and public services can perpetuate economic informality, as individuals and businesses have fewer practical opportunities to participate in the formal economy, creating a cycle of low tax revenue,[88] constrained public investment and limited economic development.[89] This creates a self reinforcing cycle as financial exclusion weakens capital markets, and then weak capital markets and

[82] Ghulam, Yaseen, et al. 'SMEs" Access to Bank Financing During the Financial Crises in Europe'. *Journal of Small Business Strategy*, vol. 35, no. 1, Jan. 2025, https://doi.org/10.53703/001c.124820.

[83] Bańkowska, Katarzyna, et al. 'Access to Finance for Small and Medium-Sized Enterprises After the Financial Crisis: Evidence From Survey Data'. *Www.Ecb.Europa.Eu*, June 2020, https://www.ecb.europa.eu/press/economic-bulletin/articles/2020/html/ecb.ebart202004_02~80dcc6a564.en.html.

[84] *MSME finance*. (n.d.). IFC. Retrieved 22 September 2026, from https://www.ifc.org/en/what-we-do/sector-expertise/financial-institutions/msme-finance

[85] *Capital Markets | World Bank Group*. (2026). Worldbank.Org. https://www.worldbank.org/ext/en/topic/financial-sector/capital-markets

[86] 'Public Services | Eurofound'. *Eurofound*, 2025, https://www.eurofound.europa.eu/en/topics/public-services.

[87] Singh, P. (2017). *Bankopedia*. Bankopedia. https://www.bankopedia.co.in/vocabulary/f/fiscal-capacity

[88] *Increasing tax revenue in developing countries*. (n.d.). World Bank Blogs. Retrieved 22 September 2026, from https://blogs.worldbank.org/en/impactevaluations/increasing-tax-revenue-developing-countries

[89] World Bank. (n.d.). *The long shadow of informality: Challenges and policies*. World Bank. Retrieved 22 September 2026, from https://www.worldbank.org/en/research/publication/informal-economy

tax systems perpetuate exclusion by failing to provide the infrastructure, oversight, capital and public goods necessary to bring marginalised populations into the formal economy.[90] And although there is an idea that digital financial technologies offer potential solutions by increasing transaction visibility and reducing compliance costs,[91] their effectiveness depends on complementary investments in digital literacy and regulatory frameworks[92] which ensure that these tools serve rather than burden low income households and small enterprises, yet the standard in developing countries for these domains is inadequate.[93]

Perpetuation of Poverty and Social Inequalities

This cycle of financial exclusion further entrenches broader societal challenges and inequalities and similarly deepens poverty, centreing on households excluded from formal savings, loans and insurance:[94] In the case of an economic shock, they are left with little options but to rely on informal lenders or to deplete their meagre assets out of necessity, which in turn diminishes these households' ability to invest in education or productive activities that could lift them out of poverty. [95]Therefore, when whole communities are locked out of public services and formal banks due to weak infrastructure, economic growth can disproportionately benefit those already participating in formal banking systems, leaving low-income households trapped in low productivity work with limited pathways to escape.

Challenges in the Deployment of Digital Financial Infrastructure in LEDCs

Historically, expanding financial access required building physical bank branches and networks in rural areas, and as a consequence was a costly and ineffective approach in developing regions hampered by weak infrastructure (unreliable electricity, poor road conditions) and informal cash economies.[96] Digital financial infrastructure can reduce some of these physical limitations by making financial services available to any mobile user, however the following impediments can surface when

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[90] Alexei Miksjuk, et al. 'Macroeconomic Challenges of Fragility and Policies for Stability and Growth'. *Departmental Papers*, vol. 2026, no. 003, Feb. 2026, https://doi.org/10.5089/9798229031479.087.A001.

[91] Brockmeyer, Anne, and Magaly Somarriba. *Electronic Payment Technology and Tax Compliance Evidence From Uruguay's Financial Inclusion Reform*. https://documents1.worldbank.org/curated/en/559201645710893209/pdf/Electronic-Payment-Technology-and-Tax-Compliance-Evidence-from-Uruguay-s-Financial-Inclusion-Reform.pdf.

[92]*Supporting informed and safe use of digital payments through Digital Financial literacy: Full report*. (2025). OECD. https://www.oecd.org/en/publications/supporting-informed-and-safe-use-of-digital-payments-through-digital-financial-literacy_21de47d1-en/full-report.html

[93]Chang, S., Duflos, E., & Chalwe-Mulenga, M. (2026, May 18). *Solutions to protect consumers from fraud in digital finance*. CGAP. https://www.cgap.org/research/publication/solutions-to-protect-consumers-fraud-in-digital-finance

[94]CGAP. (2022, September). *About financial inclusion*. Cgap.Org. https://www.cgap.org/about/financial-inclusion

[95] *The role of financial services in building household resilience*. (2016, April 22). CGAP. https://www.cgap.org/research/publication/role-of-financial-services-in-building-household-resilience

[96]Chidimma Maria-Gorretti Umeaduma. (2023). Financial inclusion strategies for poverty reduction and economic empowerment in underbanked rural populations globally. *World Journal of Advanced Research and Reviews*, *18*(1), 1263–1280. https://doi.org/10.30574/wjarr.2023.18.1.0709

an individual is overdependent on digital finance models.[97] This shift creates new challenges, particularly when individuals and economies become heavily dependent on digital financial systems. Digital services depend largely on electricity, mobile networks and internet connectivity, so where such systems are unreliable or unavailable, individuals may be unable to access their money, make payments or receive financial assistance.[98]

Weak e-KYC Frameworks

"e-KYC is only as trustworthy as the evidence used to bind a digital application to a real person."[99] What this means is that trustworthy e-KYC frameworks use standardized biometric verification, identity validation and fraud detection to ensure that a genuine user is behind the screen and not a fraudulent synthetic identity.[100] If these practices are tenuous or even missing, digital financial systems become vulnerable to identity fraud and impersonation attacks, which not only erode institutional trust as they did not reliably distinguish a real user from an impersonation,[101] therefore incurring penalties on genuine consumers and giving them incentive to regress back to informal finance, but also risk excluding legitimate users, especially the unconfident ones in rural communities, reinforcing the very barriers to inclusion that e-KYC is intended to remediate. In LEDCs, these risks can be particularly pronounced where identification systems have limited coverage and digital infrastructure is less reliable because individuals in such communities may lack official identification, reliable connectivity, or the digital skills required to complete e-KYC procedures, making it harder for financial institutions to verify them remotely. As a result, weak e-KYC can manifest as a double barrier to inclusion since inadequate verification can increase fraud yet stringent or unreliable verification can prevent legitimate users primarily from accessing formal financial services.

Weak Data Governance Regulations

Strong and reliable data governance systems require limiting data collection to what's strictly necessary, forbidding repurposing without permission and holding institutions accountable for data

[97] World Bank Group. (2025b, July). *How countries can expand access to digital financial services*. World Bank Group. World Bank. https://www.worldbank.org/en/topic/financialsector/publication/digital-financial-services

[98] Tchioffo Kodjo. (2018b). *Digital Financial inclusion - ITU PP-18 - plenipotentiary conference*. ITU Plenipotentiary Conference 2018 (PP-18). https://www.itu.int/web/pp-18/en/article/digital-financial-inclusion

[99] NHI Mgmt Group Editorial Team. 'What Happens When eKYC Is Deployed Without Strong Identity Validation and Fraud Detection?'. *NHI Management Group*, 8 Sept. 2026, https://nhimg.org/faq/what-happens-when-ekyc-is-deployed-without-strong-identity-validation-and-fraud/.

[100] NHI Mgmt Group Updated. 'E-KYC'. *NHI Management Group*, 26 Aug. 2026, https://nhimg.org/glossary/e-kyc/.

[101] Mungai, Robert. 'Synthetic Identity Fraud a Critical Primary National Security Priority'. *Social Science Research Network*, Jan. 2024, https://doi.org/10.2139/ssrn.4770398.

misuse in order to ensure that people stay in control of their digital footprint.[102] If regulations for data governance are vague or poorly enforced, as they are found to be in weak or expedient digital frameworks, storing data in systems is a liability and in developing economies, consumers frequently rely on cash or informal credit markets[103] and the promise of digital finance, to bring everyone into the formal economy, and make services cheaper and more accessible for everybody[104], is compromised under inadequate oversight.

Unaffordability of Digital technologies and lack of investment

Although DFS can reduce the cost of accessing banking, individuals still need the appropriate technology to use them, such as smartphones, SIM cards and mobile data. For low-income households, purchasing and maintaining a smartphone can be a significant expense, especially when income is already needed for essential goods and services.[105] This creates a digital divide between individuals who can afford modern technology and those who cannot, meaning that the process of digitalisation may unintentionally exclude the poorest members of society. Furthermore, to move beyond basic device ownership and unlock the full potential of these tools for genuine financial inclusion, digital financial services require significant investment in internet networks, mobile connectivity, data centres and secure payment systems, and Digital Public Infrastructure (DPIs), including digital identification and compatible payment systems that enable individuals to access and use financial services securely and efficiently.[106] However, LEDCs have limited government budgets and may struggle to attract investment as providing infrastructure in remote or rural areas can be expensive and generate low profits. Therefore, without sufficient investment, digital services may remain concentrated in urban areas which therefore leaves rural and poorer populations excluded.[107]

Lack of Cybersecurity Capacity

The expansion of digital finance means that large quantities of sensitive personal and financial information are stored electronically, including names, account details and transaction histories.[108] Many LEDCs may lack the financial resources and infrastructure, technical expertise and regulatory

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[102]'Digital Safeguards'. *Worldbank.Org*, World Bank Group, 10 Apr. 2025, https://www.worldbank.org/en/brief/2025/04/10/digital-safeguards.

[103] Bains, Parma, and Tamas Gaidosch. *Privacy Technologies & the Digital Economy*. 2025, https://doi.org/10.5089/9798229004299.001.a001.

[104] Tyson, Laura, and Susan Lund. 'The Promise of Digital Finance'. *McKinsey & Company*, 14 Dec. 2016, https://www.mckinsey.com/mgi/media-center/the-promise-of-digital-finance.

[105] "Strategies Towards Universal Smartphone Access Report – Broadband Commission." *Itu.Int*, 2025, https://broadband.itu.int/publication/strategies-towards-universal-smartphone-access/.

[106] *Digital public infrastructure | United Nations Development Programme*. (n.d.). UNDP. Retrieved 22 September 2026, from https://www.undp.org/digital/digital-public-infrastructure

[107] "How Countries Can Expand Access to Digital Financial Services." *World Bank*, https://www.worldbank.org/en/topic/financialsector/publication/digital-financial-services.

[108] Gaidosch, T., Islam, E., Khiaonarong, T., Ravikumar, R., & Wilson, C. (n.d.). *Front matter*. https://doi.org/10.5089/9798229026185.087.a000

institutions necessary to adequately protect this sensitive information.[109] Weak cybersecurity can therefore increase the risk of data leaks, identity theft, financial fraud, and cyberattacks against financial institutions which can cascade into direct financial losses[110] while also reducing public trust in digital banking because users are uncertain of whether their data is being kept safe, simply due to a capacity deficit. The risks associated with large-scale digital databases can be illustrated by India's Aadhaar system, where reports in 2018 alleged that unauthorised access had exposed personal and banking information, although the Unique Identification Authority of India (UIDAI) disputed that the central Aadhaar database itself had been breached.[111] Therefore, the Aadhaar case serves as an example of how weaknesses in cybersecurity and data governance can expose sensitive information and resuscitate the importance of institutional capacity in managing large-scale digital systems.

Weak Digital Literacy Rates among Consumers

Digital financial services require consumers to have basic digital and financial knowledge in order to use them effectively.[112] Users need to understand how to operate banking applications, make digital payments, protect their passwords and recognise fraudulent activity.[113] In areas with low digital literacy like developing states, consumers may struggle to understand these systems or may become vulnerable to scams and financial fraud.[114] This means that providing access to internet connection alone does not guarantee financial inclusion. Governments may therefore need to provide digital education and training alongside the development of digital financial infrastructure, or embed digital literacy into public networks and websites to make them comprehensive for all individuals regardless of differences in literacy levels.[115]

Timeline of Events

Date	Description of Event

[109] IMF. (2024, April). *Global financial stability report, april 2024 - chapter 2 available now; coming soon: Chapter 3 (april 9 at 1pm ET) ; chapter 1 (april 16, 10:15Am ET))*. IMF. https://www.imf.org/en/Publications/GFSR/Issues/2024/04/16/global-financial-stability-report-april-2024

[110] *Cybersecurity | digital finance inclusion*. (2016b). Worldbank.Org. https://digitalfinance.worldbank.org/topics/cybersecurity

[111] *Aadhaar data breach: What happened, impact, and lessons | Huntress*. (2015b). Huntress. https://www.huntress.com/threat-library/data-breach/aadhaar-data-breach

[112] Choung, Y., Chatterjee, S., & Pak, T.-Y. (2023). Digital financial literacy and financial well-being. *Finance Research Letters*, *58*(Part B), 104438. https://doi.org/10.1016/j.frl.2023.104438

[113] NHI Mgmt Group Editorial Team. (2026, September 8). *What happens when eKYC is deployed without strong identity validation and fraud detection?* NHI Management Group. https://nhimg.org/faq/what-happens-when-ekyc-is-deployed-without-strong-identity-validation-and-fraud/

[114] *Supporting informed and safe use of digital payments through Digital Financial literacy: Full report*. (2025b). OECD. https://www.oecd.org/en/publications/supporting-informed-and-safe-use-of-digital-payments-through-digital-financial-literacy_21de47d1-en/full-report.html

[115] *Skills for the digital economy*. (2020). ITU. https://www.itu.int:443/en/mediacentre/backgrounders/Pages/skills-development-digital-economy.aspx

February 2 - 4 1997	The first Microcredit Summit was held in Washington, DC, bringing global attention and political will to microfinance as a tool for economic empowerment, with a primary goal of reaching 100 million of the world's poorest families with essential financial services and employment by 2005.[116]
March 6, 2007	Kenya launches M-Pesa, revolutionarily shifting global financial inclusion away from physical banks towards digital financial infrastructure.[117]
2008	Mobile banking is accelerated as the use of smartphones becomes widespread.[118]
December 10, 2010	The G20 launches the Global Partnership for Financial Inclusion. [119]
July 27, 2015	Adoption of the Addis Ababa Action Agenda (AAAA), a global framework for shaping economic policies and supporting sustainable development.[120]
April 11, 2016	India launched the Unified Payment Interface (UPI), boosting fintech adoption.[121]
April 14, 2016	The World Bank launched the Identification for Development (ID4D) initiative, helping all people access

[116]Christensen, Marcie. 'Women's Microfinance Initiative - Together Women Rise'. *Together Women Rise* -, 15 Nov. 2014, https://togetherwomenrise.org/foodforthought/womens-microfinance-initiative-2/.

[117]'M-Pesa: Kenya's Mobile Wallet Revolution'. *BBC News*, 22 Nov. 2010, https://www.bbc.com/news/business-11793290.

[118] *The history of digital banking | UBS Switzerland*. (2022, September 5). Women's Wealth Academy. https://www.ubs.com/ch/en/wealth-management/womens-wealth/magazine/articles/history-of-digital-banking.html

[119]*About GPFI | gpfi*. (n.d.). Www.Gpfi.Org. Retrieved 21 September 2026, from https://www.gpfi.org/about-gpfi

[120]*Financing for sustainable development*. (n.d.). Www.Un.Org. Retrieved 21 September 2026, from https://www.un.org/esa/ffd/publications/aaaa-outcome.html

[121]MC, A., & Shanmugam, K. (2023). Unified payment interface—Taking India to the next generation in payments. *Journal of Information Technology Teaching Cases*, *14*(2), 241–246. https://doi.org/10.1177/20438869231178843

	financial services through trusted and official digital identification systems.[122]
January 3, 2018	The Aadhaar system data breach happens, posing a significant security threat to many users of this digital financial service.[123]
March 11, 2020	Official declaration of a global pandemic by the World Health Organization (WHO) because of the virus COVID-19, bringing about acceleration of digital banking and payment systems.[124]
2023	The World Bank advanced Digital Public Infrastructure (DPI) as a cornerstone of inclusive development through its ID4D and GP2x initiatives.[125]

Major Actors Involved

Kenya

Kenya is currently one of the most successful examples of mobile money adoption among LEDCs. Mobile money transfer has overtaken informal financial groups, with 61% of users in rural towns being registered mobile money transfer users, 51% still operate in informal financial groups, in other words, unregulated and community-based networks, operating outside traditional banking models where groups of people pool money together to save, spend or invest, and approximately 36% using banks.[126] Following the launch of M-Pesa in 2007, digital transactions have driven socio-economic growth by making financial services accessible to anyone. M-Pesa has improved financial inclusion in Kenya by giving rural and low-income communities easier access to financial services and has also increased economic activity by making transactions quicker and cheaper, while allowing families to

[122] *Identification for development strategic framework*. (2016). https://thedocs.worldbank.org/en/doc/21571460567481655-0190022016/render/April2016ID4DStrategicRoadmapID4D.pdf

[123] *Aadhaar data breach: What happened, impact, and lessons | Huntress*. (2015). Huntress. https://www.huntress.com/threat-library/data-breach/aadhaar-data-breach

[124] Kosse, A. (2021, December 9). *Covid-19 accelerated the digitalisation of payments*. Bank for International Settlements. https://www.bis.org/publications/202607-commentary-red-book

[125] Group, W. B. (2023, October 16). *Creating Digital Public infrastructure for empowerment, inclusion, and resilience*. World Bank Group. World Bank. https://www.worldbank.org/en/results/2023/10/12/creating-digital-public-infrastructure-for-empowerment-inclusion-and-resilience

[126] *How does mobile money in Kenya affect financial-inclusion A majorthdatabase mentioninginatoryal inclusion?* (2012, April 1,9). CGAP. https://www.cgap.org/blog/how-does-mobile-money-in-kenya-affect-financial-inclusion

send and receive money more easily.[127] Data showed that access to M-Pesa in 2016 contributed to wider socio-economic development and helped lift approximately 194,000 Kenyan households out of poverty.[128]

India

Although low formal banking access was a major alarm point in 2013 when data revealed that a substantially smaller share of adults held financial accounts, over a decade later, with the largest digital biometric identity and real-time payment systems being built, that number had reached 85% of the country's population.[129] An important building block for reaching 'Digital India' is the Aadhaar digital identity system, which provides residents with a unique digital identity, although errors, exclusion and identity misuse remain possible.[130] Something worth mentioning is also the 2018 allegations concerning unauthorised access to Aadhaar-related information, which were disputed by the authority responsible for the central database, especially to a database that has over 1.4 billion users.[131] India's payment system called the Unified Payments Interface (UPI) helped enable rapid and secure money transfers between people, banks and governments. The UPI system enables bank-account-based payments through participating applications and is regulated by the Reserve Bank of India (RBI).[132] India's experience is proof that with such systems and its significant progress in developing public infrastructure preserves the right of all states to economic participation.

United States of America

The US fintech market is a global leader valued at over $60 billion,[133] with centres in New York and San Francisco. US financial technology firms, such as PayPal, Visa, and Mastercard, provide innovative services and foster a more equitable, financially-secure economy through the integration

[127] Demombynes, Gabriel,Thegeya, Aaron Miruri. (n.d.). *Kenya's mobile revolution and the promise of mobile savings*. World Bank. Retrieved 21 September 2026, from https://documents.worldbank.org/en/publication/documents-reports/documentdetail/900911468047101453

[128] Suri, T., & Jack, W. (2016). The long-run poverty and gender impacts of mobile money. *Science*, *354*(6317), 1288–1292. https://doi.org/10.1126/science.aah530

[129] *How India leapfrogged financial inclusion*. (2026, July 9). Harvard Kennedy School. https://www.hks.harvard.edu/centers/mrcbg/programs/growthpolicy/how-india-leapfrogged-financial-inclusion

[130] Nilekani, N. (n.d.). *India's aadhaar system: Bringing E-Government to life*. https://chandlergovernance.com/governancematters/indias-aadhaar-system-bringing-e-government-to-life

[131] *Aadhaar Data breach: What happened, impact, and lessons | Huntress*. (n.d.). Huntress. https://www.huntress.com/threat-library/data-breach/aadhaar-data-breach

[132] *What is Unified Payments Interface (UPI)? | Stripe*. (2025, August 11). https://stripe.com/en-gr/resources/more/unified-payments-interface-upi

[133] Mordor Intelligence. (2023). *US fintech market | growth, trends, forecast (2019 - 2024)*. Www.Mordorintelligence.Com. https://www.mordorintelligence.com/industry-reports/us-fintech-market

of digital assets and innovative technology in traditional financial systems.[134] The USA is considered an actor that can significantly help with this issue, as they already have an allocated body that is established to deliver civilian foreign aid and development assistance, the United States Agency for International Development (USAID)[135]. Regarding the issue of financial inclusion solely, projects have been approved and funded by the U.S. Development Finance Corporation (DFC), like the $300 million investment aimed toward advancing inclusive economic development in Brazil.[136] The USA also supports digital public infrastructure, including but not limited to Digital Identity, Digital Payment, and Data Exchange, through partnerships with institutions such as the World Bank Group (WBG), the International Monetary Fund (IMF), the G20, and regional development banks.[137] Driven by the rapid growth of fintech firms, the US promoted financial innovation and regulatory best practices, reducing unnecessary barriers[138], but also encouraging private sector investment as key to modernising the country's infrastructure and boosting economic growth.

World Bank Group (WBG)

The World Bank Group (WBG) is a global partnership of five institutions working to reduce poverty and foster shared prosperity in developing countries.[139] This family of organisations is arguably the most important international change-maker regarding financial inclusion, providing many programmes that support the concept of 'Open Data implementation', which refers to a systematic process where governmental and institutional data are freely available to the public without restrictions on reuse,[140] and utilise this to address specific challenges. [141] The World Bank Group collabourated with governments, regulators, and private-sector actors for evidence-based policies and regulations regarding financial inclusion, while also modernising financial infrastructure, like

[134] Seedtable. (n.d.). *Best Fintech Startups in United States (2026) | Seedtable*. https://seedtable.com/best-fintech-startups-in-united-states

[135] PSI. (2026, May 7). *United States Agency for International Development - PSI*. PSI - Nonprofit Organization Making It Easier for People in the Developing World to Lead Healthier Lives and Plan the Families They Desire by Marketing Affordable Products and Services. https://www.psi.org/partner-with-us-archived/governments-and-foundations/united-states-agency-for-international-development-usaid/isPayment\

[136] *DFC approves new investments to advance sustainable economic development, respond to COVID-19, and advance gender equity*. (2021, including butMarch 9). DFC. https://www.dfc.gov/media/press-releases/dfc-approves-new-investments-advance-sustainable-economic-development-respond

[137] Federation of American Scientists. (2025, December 8). *Digital public infrastructure should be America's next public works project*. https://fas.org/publication/digital-public-infrastructure-act/

[138] The White House. (2026, June 22). *Integrating Financial Technology Innovation into Regulatory Frameworks*. https://www.whitehouse.gov/presidential-actions/2026/05/integrating-financial-technology-innovation-into-regulatory-frameworks/

[139] World Bank Group. 'About Us'. *World Bank Group*, 2024, https://www.worldbank.org/ext/en/who-we-are.

[140] 'What Is Open Data | Data.Europa.Eu'. *Data.Europa.Eu*, https://data.europa.eu/en/dataeuropa-academy/what-open-data.

[141] *Open Data Toolkit |Technical Assistance and Funding | Data*. (n.d.). World Bank. https://opendatatoolkit.worldbank.org/en/data/opendatatoolkit/technical-assistance

expanding digital payment systems and broadening the use of digital identification.[142] A key component of the work undertaken by the WBG has been the development of the necessary infrastructure for achieving financial inclusion, such as the launch of the ID4D project. In doing so, the WBG has helped countries build inclusive and secure digital identification systems, which could aid in accessing various banking facilities including bank accounts, mobile money, payments from the government, and other forms of formal financial service offerings.[143] In addition, the WBG has created the Government-to-Person Payments (G2Px) which has helped in digitizing payments made by governments including payments related to social benefits in a cost-effective manner while enabling people who previously had limited access to formal financial systems to join the system. [144]Another important innovation of the WBG is the Global Findex Database in 2011, as it is a major global source of data that examines the way in which adults around the world use financial services across 141 economies.[145] Therefore it is clear that the WBG has been concerned with building the Digital Public Infrastructure (DPI) required for achieving inclusion, as opposed to just increasing access to financial products for LEDCs.

International Monetary Fund (IMF)

The International Monetary Fund (IMF), established in 1944, is an international organization with 191 member states with the mission of encouraging international monetary cooperation, assessing the economic policies of its member countries, monitoring the global economy, and providing financial assistance.[146] The IMF is notably relevant to expanding financial inclusion as it helps developing countries identify economic risks while supporting the safe integration of digital financial services. In this context, it monitors fintech developments and provides policy advice and assistance to strengthen financial-sector resilience, consumer protection and cybersecurity.[147] Furthermore, the IMF contributes to discussions surrounding cross-border payments and digital finance, offering support to countries developing systems that are both more accessible and secure.[148] The organisation also supports LEDCs in expanding digital financial infrastructure while ensuring that greater access

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[142] *Financial inclusion*. (n.d.). World Bank Group. https://www.worldbank.org/ext/en/topic/financialsector/financial-inclusion#tab-strategy

[143] *About us | identification for development*. (2018). Worldbank.Org. https://id4d.worldbank.org/about-us

[144] *Government-to-person (G2Px)*. (n.d.). World Bank. Retrieved 22 September 2026, from https://www.worldbank.org/en/programs/g2px

[145] *The Global Findex database*. (n.d.). Data | FinDev Gateway. https://www.findevgateway.org/data/globalfindex-database

[146] *What Is the International Monetary Fund? Core functions | BBVA CIB*. (2026, July 9). BBVA CIB. https://www.bbvacib.com/insights/news/investment-banking-what-is-international-monetary-fund-imf/

[147] Menon, A. (2024, June 28). IMF's role and impact on the global economy. *TRADEJINI*. https://www.tradejini.com/blogs/the-role-and-impact-of-the-international-monetary-fund-in-the-globaleconomgovernmentsy

[148]IMF. (2025, January 29). *Digital payments and finance*. IMF. https://www.imf.org/en/Topics/digital-payments-and-finance

to financial services contributes to sustainable economic development, without creating additional financial risks.[149]

Alliance for Financial Inclusion (AFI)

The Alliance for Financial Inclusion (AFI) is a policy leadership alliance and a network of central banks and financial regulators from developing and emerging economies (LEDCs), intending to help governments with knowledge-sharing and developing financial inclusion policies, so that everyone can benefit from financial services. With 1 billion people financially included and 89 member institutions[150], AFI has worked on the sector of consumer protection by proposing policy models[151], on Small and Medium-Sized Enterprise (SME) finance by creating an SME Finance Working Group in 2013 (SMEFWG)[152], and actively empowering consumers by promoting policies and investing in those who face significant barriers in accessing and using financial services, whether the gap is caused by gender or financial illiteracy.[153] With the Alliance for Financial Inclusion and the knowledge-sharing policies they have proposed, LEDCs are not dependent on policies created by wealthier countries but learn from one another.

UN Documents and International Agreements

Addis Ababa Action Agenda (AAAA)[154]

Adopted by the United Nations General Assembly on July 27th, 2015, the Action Agenda focuses on providing a global framework to foster sustainable development and eliminate gaps that hinder the economic growth of nations. It highlights the importance of financial access and emphasizes the need for marginalised communities, such as women and rural populations, to have access to financial services, as well as financial literacy, advocating to support this cause through legislation and regulation, to encourage microfinance institutions and the use of innovative tools, but also to expand

[149] *The rise of public and private digital money: A strategy to continue delivering on the IMF's mandate*. (n.d.). IMF. Retrieved 21 September 2026, from https://www.imf.org/en/Publications/Policy-Papers/Issues/2021/07/28/The-Rise-of-Public-and-Private-Digital-Money-462919

[150] *AFI (Alliance for FInancial Inclusion) | ADA - Appui au développement autonome*. (n.d.). https://adaimpact.lu/en/about-ada/ada-partners/afi-alliance-financial-inclusion

[151] Vasnani, H., & Vasnani, H. (2026, July 5). AFI consultation: Policy Model for Consumer Protection - Alliance for Financial Inclusion. *Alliance for Financial Inclusion - Bringing smart policies to life*. https://afi-global.org/news/afi-consultation-policy-model-for-consumer-protection/

[152] Alliance for Financial Inclusion. (2026, May 22). *SME Finance Working Group (SMEFWG) - Alliance for Financial Inclusion*. Alliance for Financial Inclusion - Bringing Smart Policies to Life. https://afi-global.org/what-we-offer/working-groups/sme-finance-working-group-smefwg/

[153] Alliance for Financial Inclusion. (2026, July 7). *Gender Inclusive Finance - Alliance for Financial Inclusion*. Alliance for Financial Inclusion - Bringing Smart Policies to Life. https://afi-global.org/what-we-do/gender-inclusive-finance

[154] *Document Viewer*. (n.d.). https://docs.un.org/en/A/res/69/313

peer learning through collabourations with the AFI.[155] While these national strategies present a compelling vision on paper, translating the AAAA into tangible outcomes reveals various structural limitations. When assessing its practical applicability, there are gaps and unanswered questions about funding and the extent to which the Agenda can effectively support LEDCs,[156] and the fact that it is not legally binding, thus its success relies on voluntary compliance.

2030 Agenda for Sustainable Development (2015)[157]

The 2030 Agenda for Sustainable Development created a plan of action consisting of 17 Sustainable Development Goals, many of which are linked to financial inclusion. SDG 8.10 calls for strengthening the capacity of domestic financial institutions to increase access to banking services.[158] Goal 9, Target 9.c aims to significantly increase access to Information and Communication Technologies (ICT), and strive to provide universal and affordable access to the internet in the Least Developed Countries (LDCs) by 2020.[159] Although this Agenda has successfully recognised financial inclusion as a key component of sustainable development, it is important to acknowledge that the SDGs are global objectives and are not legally binding and do not guarantee funding for countries to achieve the necessary investments.

Pact for the Future and Global Digital Compact (2024)[160]

The United Nations General Assembly adopted the Pact for the Future, including the Global Digital Compact, on 22 September 2024 during the Summit of the Future process. It aims to redefine international cooperation and, as stated in the agenda, is committed to five key objectives: closing digital divides and accelerating the SDGs, expanding inclusive access to the digital economy, fostering a safe and rights-based digital environment, developing data governance, and enhancing AI governance.[161] This agenda is extremely relevant to our topic, as financial inclusion is increasingly reliant on digital infrastructure, secure online systems, and equal access to digital technologies. However, while this provides a comprehensive global framework, it is highly dependent on voluntary implementation and how willing the Member States are to turn the pact into concrete policies.

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[155] Alliance for Financial Inclusion. (2024, October 31). *Addis Ababa Action Agenda - Alliance for Financial Inclusion*. Alliance for Financial Inclusion - Bringing Smart Policies to Life. https://afi-global.org/publication/addis-ababa-action-agen.da/

[156] Toth, I. (2015). The Applicability of the Addis Ababa Action Agenda in Fragile Contexts—A Reflection. *Journal of Peacebuilding & Development*, *10*(3), 116–123. https://doi.org/10.1080/15423166.2015.1105667

[157] *Document Viewer*. (n.d.). https://docs.un.org/en/A/res/70/1

[158] *SDGs | KnowSDGs*. (n.d.). https://know-sdgs.jrc.ec.europa.eu/sdg/8

[159] *Goal 9. Infrastructure, Industrialization, Innovation*. (n.d.). ITU. https://www.itu.int/en/sustainable-world/Pages/goal9.aspx

[160] *Home | Pact for the future*. (n.d.). https://www.un.org/pact-for-the-future/en

[161] Communications. (2025, July 9). *Building a future that works for all: the Pact for the Future and the Global Digital Compact*. Soroptimist International. https://www.soroptimistinternational.org/2025/06/06/building-a-future-that-works-for-all-the-pact-for-the-future-and-the-global-digital-compact/

Previous Attempts to Solve the Issue

World Bank Identification for Development (ID4D) initiative

The WBG, through its Identification for Development (ID4D) initiative, has supported financial inclusion through funding, technical assistance, and digital public infrastructure. Through its Identification for Development (ID4D) initiative, the WBG aims to help countries build inclusive and trusted identification systems by providing technical and financial assistance, conducting country assessments, and developing global guidance and standards, with its work spanning over 60 countries and 550 million people across areas including financial inclusion, digital development, and social protection,[162] however this figure demonstrates the scale of the initiative rather than necessarily measuring its impact on financial inclusion. Through collabourations with countries, the WBG oversees that identification systems are accessible and non-discriminatory, impacting more than 550 million people.[163] Unfortunately, this act faces many barriers regarding its implementation, particularly in fragile states and rural areas. ID access can be uneven across Africa, for instance, due to poor digital connectivity and the lack of fundamental records that are needed to qualify for an ID.[164] As previously mentioned, people without birth certificates or other basic legal documents may struggle to prove their identity and obtain an ID, weak data protection can create risks of personal information being misused or stolen[165] and different government agencies may use separate systems that cannot easily share identification information.[166] It is also worth noting that the initiative's success ultimately depends on the governments implementing its measures, meaning that the WBG cannot guarantee the same level of effectiveness across different countries. This is particularly relevant in states with weaker institutional capacity or limited digital infrastructure, but nevertheless, ID4D has clear merit as it provides countries with technical assistance, financing, and guidance to develop identification systems that are more inclusive and secure.

Western Balkans Investment Framework (WBIF)

The Western Balkans Investment Framework (WBIF) is a joint EU financial platform that supports infrastructure and economic development in the Western Balkans. It combines EU grants, loans from international financial institutions, and national funding to finance projects in areas such as transport,

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[162] *About us | identification for development*. (2018). Worldbank.Org. https://id4d.worldbank.org/about-us

[163] *Home | Identification for Development*. (n.d.). https://onecovered,id4d.worldbank.org/

[164] *The importance of digital national ID in Africa*. (2024). World Bank Blogs. https://blogs.worldbank.org/en/digital-development/the-importance-of-id-access-in-three-charts--insights-from-sub-s

[165]Research. (2024, September 9). *No identity, no protection: How lack of identity documents drives exploitation and modern slavery | freedom fund*. Freedom Fund. https://www.freedomfund.org/news/no-identity-no-protection/

[166]*The data disconnect: Tech barriers in public sector data sharing | capita*. (2025). Capita Corporate. https://www.capita.com/news-and-insights/insights/2025/data-disconnect-addressing-technical-barriers-effective-public-sector-data-sharing

energy and digital infrastructure.[167] The WBIF has provided over €145 million in technical assistance, including €15 million specifically for broadband and digital infrastructure planning and digital connectivity in remote and lower-income communities, thus strengthening Digital Public Infrastructure in the region,[168] and has supported small and medium-sized enterprises (SMEs) in integrating them into digital finance across its private sector interventions, prioritising start-ups, rural MSMEs, and those operating in the manufacturing sector.[169] Its digital projects aim to make reliable internet and digital services more accessible, allowing people in rural areas to access online banking, digital payments and other financial services. However, differences in national regulations and priorities can make regional cooperation more difficult and expanding digital infrastructure increases the need for strong cybersecurity and data protection systems. Conflicting priorities is a central and recurring theme within WBIF dialogues because the WBIF frequently selects initiatives before detailed planning permissions and technical designs are entirely ready to be executed,[170] leading to implementation delays, which then had cascading effects on the speed of possible integration.

The Global Partnership for Financial Inclusion (GPFI)

The GPFI was created by the G20 in 2010 to strengthen the global financial inclusion agenda by bringing together governments, international organisations, and other stakeholders to improve access to and use of formal financial services, particularly for underserved populations. More specifically, it became the main mechanism for implementing the G20 Financial Inclusion Action Plan, using policy coordination, knowledge sharing, data collection and peer learning[171] to help countries develop and measure their own financial inclusion strategies. Its work helped institutionalise financial inclusion within the global development agenda and supported countries in developing policies around consumer protection, financial literacy, and access to financial services.[172] Its impact can be seen in the wider expansion of financial access, with global account ownership increasing from 51% in 2011 to 76% in 2021,[173] although this progress cannot be attributed to the GPFI alone. However, its influence is primarily based on coordination and policy guidance[174] without binding mechanisms and interpretation, which is particularly noticeable for rural populations in developing countries, where limited infrastructure and access to financial providers may obstruct the implementation of

[167] "About the WBIF." *Www.Wbif.Eu*, https://www.wbif.eu/about/about-wbif.

[168] *Private sector | WBIF EU support for MSMEs & innovation*. (2025). Wbif.Eu. https://www.wbif.eu/sectors/private-sector

[169] *Homepage - European Union and the Western Balkans*. (2026). Wbif.Eu. https://www.wbif.eu

[170] *Special report 16/2026: Western balkans investment framework*. (2026). European Court of Auditors. https://www.eca.europa.eu:443/en/publications?ref=SR-2026-16

[171] *About GPFI | gpfi*. (2026). Gpfi.Org. https://www.gpfi.org/index.php/about-gpfi

[172] *Our work | GPFI*. (2017). Gpfi.Org. https://www.gpfi.org/our-work

[173] World Bank Group. (2023, August 11). *Global FINDEX database 2021 survey headline findings on account ownership*. World Bank Group. World Bank. https://www.worldbank.org/en/publication/globalfindex/brief/the-global-findex-database-2021-chapter-1-ownership-of-accounts?utm

[174] *G20 2023 financial inclusion action plan | GPFI*. (2023, September 19). Gpfi.Org. https://www.gpfi.org/publications/g20-2023-financial-inclusion-action-plan

financial inclusion policies at the national level in communities that are most affected by exclusion.[175] The GPFI therefore may be viewed as an important step towards concerted global action, however, it is particularly impactful when international guidance is paired with domestic resources.[176]

M-Pesa Initiative

M-Pesa is a mobile money service launched in Kenya in 2007 by Safaricom and Vodafone. It allows users to send and receive money, make payments, save money and access financial services using a mobile phone, without necessarily needing a traditional bank account.[177] The World Bank found that mobile banking in Kenya helped reduce the vitality of distance from physical bank branches when accessing financial services. M-pesa's relevance lies in demonstrating the potential of mobile money in shaping digital financial inclusion, as its widespread adoption in Kenya has reduced transaction costs and enables underserved populations to interact with digital payments and other financial services.[178] Its success, however, is difficult to replicate universally. Kenya benefited from extensive mobile coverage and an enabling regulatory environment,[179] in comparison to countries with weaker infrastructure or less supportive regulation, such as LEDCs, who may not achieve the same level of impact.[180] Thus, M-Pesa demonstrates the potential of mobile finance to expand formal financial access, its importance lies more in demonstrating what can be achieved when digital infrastructure and regulation operate together than in providing a universally replicable solution to financial exclusion.

Possible Solutions

Improved Connectivity and Physical Infrastructure

Governments should collabourate with international financial institutions, such as but not limited to the World Bank, private telecommunications companies, and FinTech firms to expand mobile networks, reliable electricity, and affordable internet access throughout underserved rural areas. As

[175]Group, W. B. (2017, November 7). *Financial access*. World Bank Group. World Bank. https://www.worldbank.org/en/publication/gfdr/gfdr-2016/background/financial-access?utm_

[176] *How to make financial sector development strategies work: Successes and failures five key lessons learned lessons learned series*. (2015). https://documents1.worldbank.org/curated/en/780001468190747225/pdf/96910-BRI-REPLACEMENT-PUBLIC.pdf?utm_

[177] King, Michael. "Is Mobile Banking Breaking the Tyranny of Distance to Bank Infrastructure? Evidence From Kenya." *World Bank Blogs*, World Bank Group, 27 Nov. 2012, https://blogs.worldbank.org/en/allaboutfinance/is-mobile-banking-breaking-the-tyranny-of-distance-to-bank-infrastructure.

[178]*Timing | digital finance inclusion*. (2018). Worldbank.Org. https://digitalfinance.worldbank.org/topics/competition/timing

[179]*Structure, framework and design of innovation facilitators*. (2018). Worldbank.Org. https://digitalfinance.worldbank.org/topics/innovation-facilitators

[180]Johnson, S. (2013, February 21). *How does mobile money in Kenya affect financial inclusion?* CGAP. https://www.cgap.org/blog/how-does-mobile-money-in-kenya-affect-financial-inclusion

mentioned, this would be achieved through government investment and public-private partnerships (PPP). With PPPs the financial costs are shared, as well as technical expertise, reducing the cost of infrastructure projects and accelerating their implementation. Throughout this project, highly excluded regions with weak digital connectivity should be prioritised, ensuring that the investments reach the populations most in need. Concluding, this addresses one of the principal barriers to financial inclusion, and as a result there will be greater participation in the formal financial system, promoting inclusive economic growth, while simultaneously limiting regional inequalities.

Creation of Regulatory Sandboxes

The creation of regulatory sandboxes, established under the supervision of relevant and competent local independent financial oversight authorities and with the aid of intergovernmental financial organisations, can help expand financial inclusion by allowing developing states to safely test new digital financial services before introducing them on a large scale, allowing for the acceleration of fintech innovation in LEDCs. Governments can create controlled environments where banks and other providers can test technologies such as digital payments and mobile banking before these services are implemented. By testing these technologies it reduces the risk of data being leaked and protects the consumer. Regulatory sandboxes can therefore encourage innovation while making digital financial services safer and more accessible.

Improving Financial and Digital Literacy

By improving financial and digital literacy it can help expand financial inclusion by teaching people how to safely use digital financial services such as mobile banking, digital payments and online banking platforms. Governments and schools can provide training programmes, particularly in low-income communities to help broaden people's understanding of budgeting, saving and online payments. It can also help individuals make better financial decisions and become more independent in managing their money. Improving financial and digital literacy would therefore help more people participate safely through the reduced risks of fraud and financial mistakes in the digital economy and benefit from financial services.

Further Reading

What Is KYC (Know Your Customer?)

Digital ID to Enhance Financial Inclusion

Digital Public Infrastructure for Inclusive Finance

Mobile Money Innovations for Financial Inclusion

Blockchain Computing for Digital Financial Inclusion

Cloud Computing Can Be a Key Enabler of Financial Inclusion

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